Research Portfolio.

Bomi Kwon

https://kwonbomi.github.io/

bomi9970@sookmyung.ac.kr

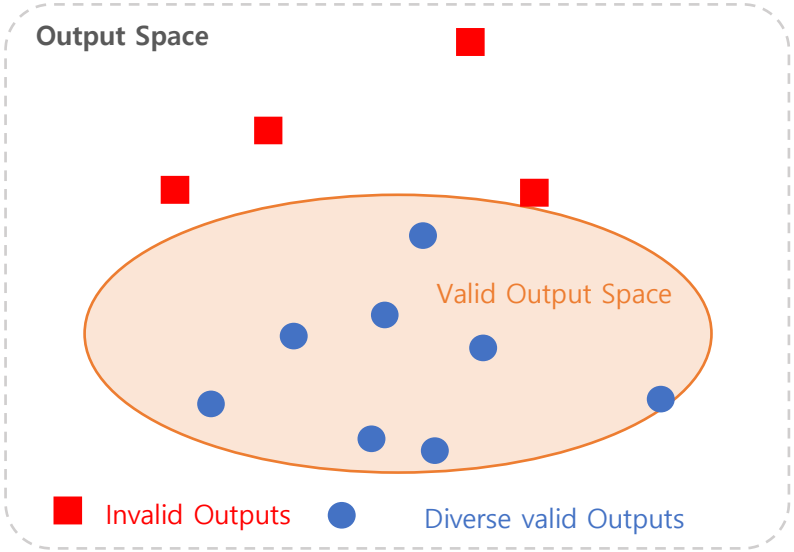
Research Focus

| Structured & Constraint-Aware Generative Modeling

Building generative models that **preserve what must remain true** while **remaining flexible** about what can vary.

Example – Humanoid Pose Generation :
joint limits and self-collision constraints must hold, while multiple text-aligned poses can remain valid

- Diffusion Models
- Structured Generation
- Constraint-Aware Generation
- Controllable Generation



High-level Human Intent / Specification
"Generate…" , "Simulate…" , "Satisfy…"

↓

Conditioned Generative Model

↓

Validity & Flexibility

1) What must remain true?

- Semantic intent
- Relational structure
- Temporal order
- Physical constraints

2) What can vary?

- Realization
- Configuration
- Low-level details
- Diverse valid outputs

How My Research Led Me Here

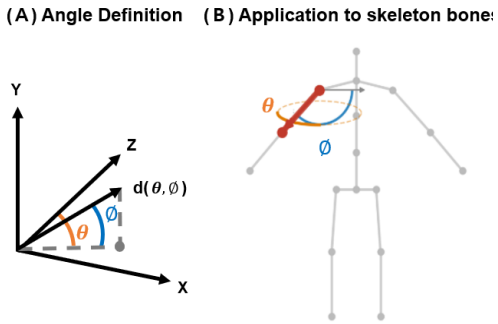
| A Diffusion-Based Trajectory toward Structured & Constraint-Aware Generation

Directional-Angle Diffusion

Represent Structure

"Generate a sitting pose."

$[\theta_1, \emptyset_1, \theta_2, \emptyset_2, \cdots]$
22D directional angles



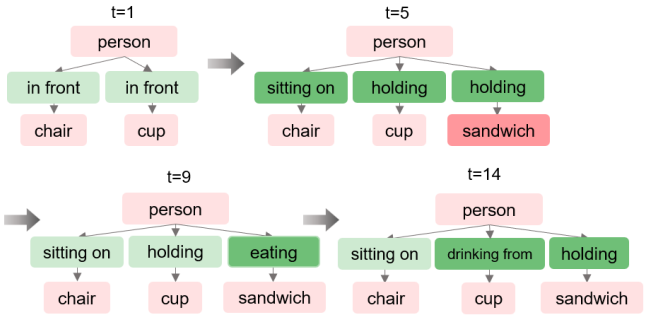
Text-Driven 2D Pose Generation
Structure-aware directional representation

TKIPS 2026

Discrete Diffusion

Generate Structure

"A person sits on the chair eats a sandwich and drinks from a cup."



Text-to-Scene-Graph Sequence Generation
Discrete relational structures over time

CIKM 2026 Accepted Short Paper

Constraint-Guided Diffusion

Constrain Generation

"Left arm stretched fully overhead"

"Arms extended out to the side, wide stance"



Text-Conditioned Humanoid Pose Generation
Joint-limit constraints & self-collision guidance

Ongoing Research

NEXT

Generalize across Structures & Constraints

Specifications as Conditions

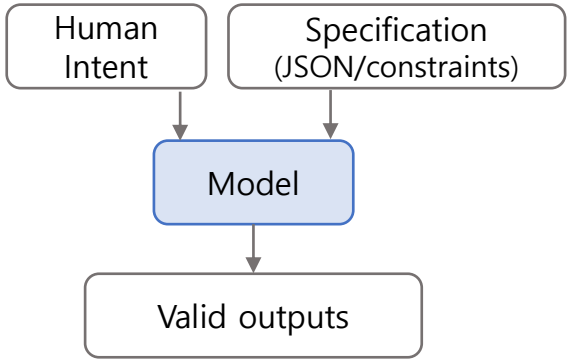
Adapt generation to new structures, embodiments, and constraints

Future Research Plan

Core Research Question
How can we build generative models that are genuinely useful for real-world problems across different domains?

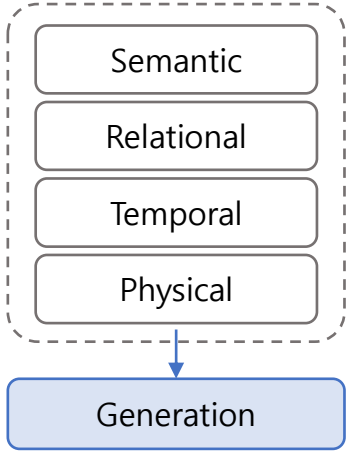
My focus is on making generative models adaptable, constraint-aware, and verifiable
– moving beyond plausible outputs toward practical use.

1. Adaptable Generation



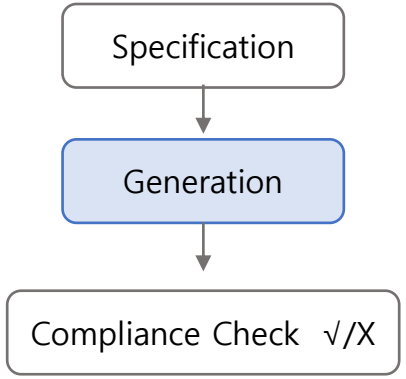
Build generative models that can adapt to different structures, domains, and constraints.

2. Constraint-Aware Generation



Satisfy task-specific requirements while preserving diversity.

3. Verifiable Generation



Explicitly evaluate whether the requested conditions are satisfied.

Long-Term Goal
Develop broadly applicable generative models that can be reliably used to solve real-world problems across different domains.

Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion

Bomi Kwon, Ki Yong Lee

| Generating time-ordered human-object interaction graphs directly from natural-language descriptions

Motivation :

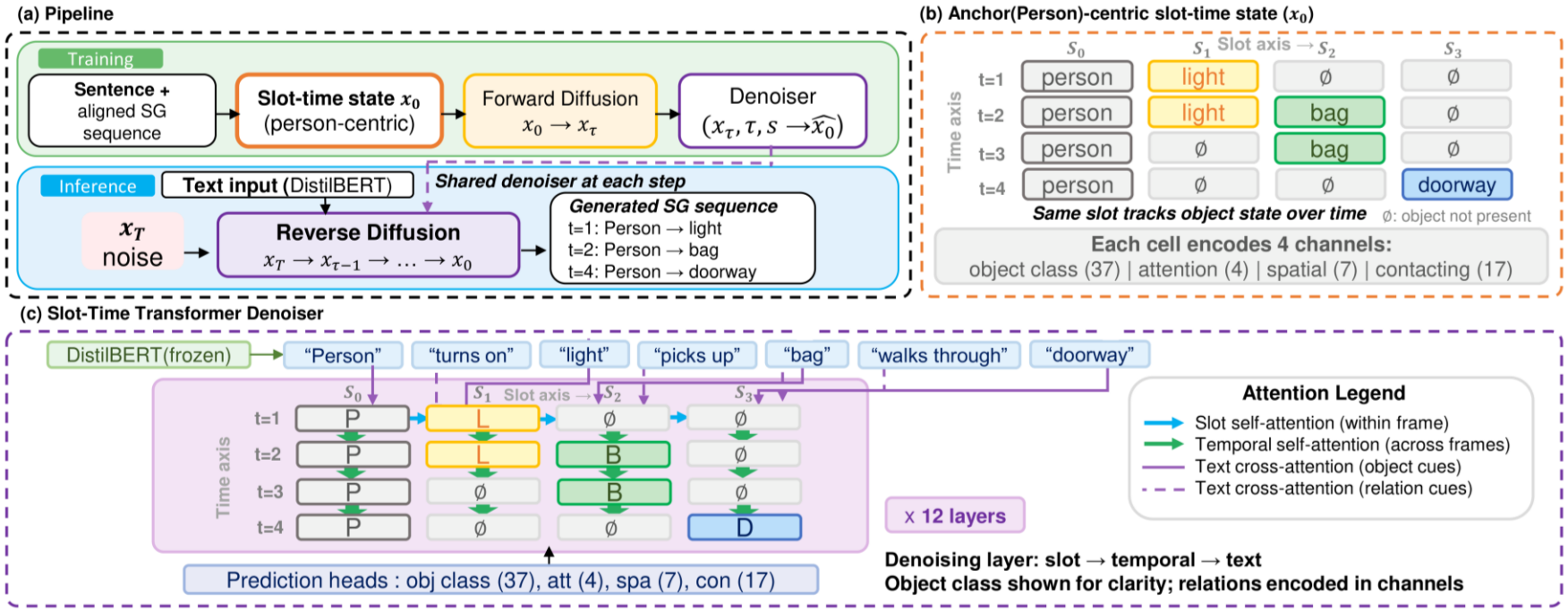
→ Existing Video-based scene graph sequence methods mainly rely on visual observations.
→ Static text-to-scene graph methods cannot represent how objects and relations evolve over time.

Research Question :

→ Can we generate a temporally evolving structured representation directly from language?

Key Idea :

1. **Anchor-centric fixed-slot representation**
   : Each slot tracks the same object across time.
2. **Discrete diffusion over categorical graph states**
   : Iterative denoising of categorical object/relation states.
3. **Slot-Time Transformer Denoiser**
   : Slot → Temporal → Text Attention



Contribution

Formulated text-to-scene-graph-sequence generation as probabilistic discrete structured generation with an anchor-centric slot-time representation.

Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion

Bomi Kwon, Ki Yong Lee

| Results & Analysis

1. Better Text-Graph Alignment

Metric	T5-base	Ours-24M	Δ
TG-Obj ↑	0.787	**0.882**	+0.095
TG-Rel ↑	0.464	**0.516**	+0.052

*TG-Obj / TG-Rel measure text-grounded object and relation alignment in the generated sequence.

→ Better alignment with ~9 x fewer trainable parameters

2. Temporal Dynamics

Metric	T5-base	Ours-144M
SK ↑ (Main temporal metric)	0.432	0.978
Appear ↑	0.000	0.996
Vanish ↑	0.000	0.996
Change-only ↑	0.000	0.888

*Temporal metrics evaluate whether object states are preserved, appear, disappear, and change correctly over time.

→ T5 fails to model explicit state transitions, while our model captures appearance, disappearance, and object changes.

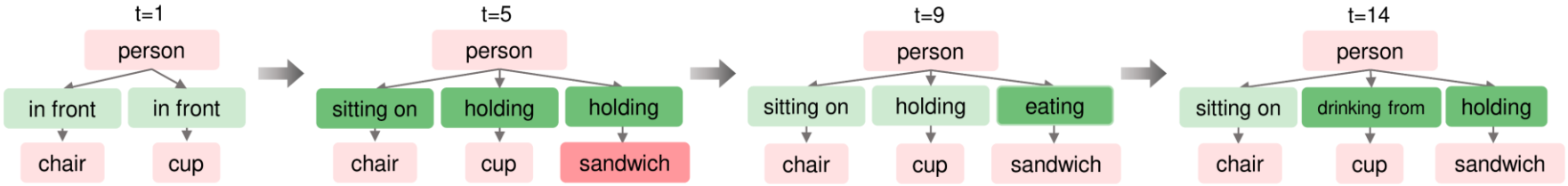
3. Why Diffusion?

Model	TG-Rel	SK	Rel	Diversity
GraphSeq-VAE	0.480	0.981	0.289	0.060
Mask-Graph	0.460	0.969	0.278	0.000
Ours-24M	0.516	0.982	0.305	0.359

→ Best **accuracy-diversity** trade-off among accurate 24M generators
→ **CFG :** better text conditioning (TG-obj 0.783 → 0.850)
→ **Stochastic sampling** : better relation recovery + diversity

4. Qualitative Result

"Person sits on the chair eats a sandwich and drinks from a cup."



The generated sequence captures how objects and relations evolve over the described event.

Text-Conditioned Humanoid Pose Generation with Physical Constraints

| Generating diverse humanoid poses from language while explicitly respecting robot-specific physical constraints

Motivation :

→ Text-conditioned generative models can produce plausible humanoid poses that violate physical constraints.
→ For robot embodiments, validity depends on embodiment-specific joint limits and self-collision constraints.

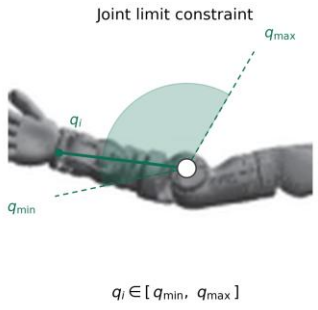
Research Question :

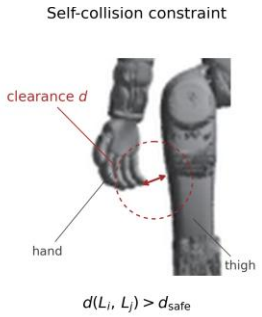
→ Can diffusion generate diverse text-aligned humanoid poses while explicitly satisfying physical constraints?

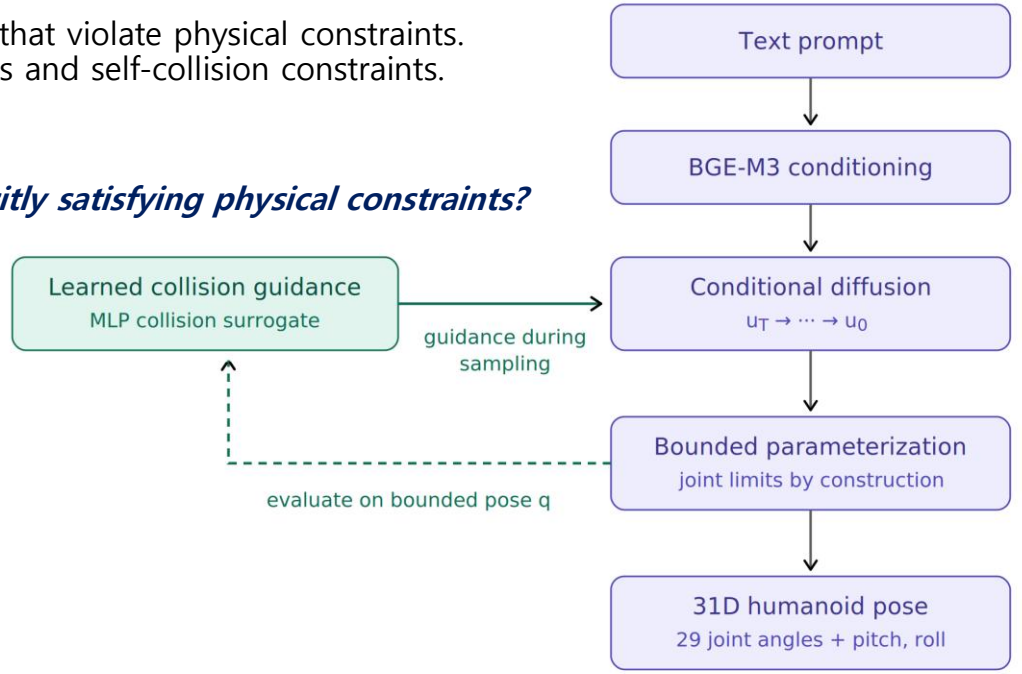
Two Key Constraints :

1. **Joint Limits**
   Each joint must remain within the valid range defined by the robot specification.

2. **Self-Collision**
   Generated body configurations should avoid interpenetration







Bounded parameterization
Joint limits built into the generation space
$q = q_{min} + (q_{max} - q_{min}) \cdot \sigma(u)$
0% violations by construction

Learned collision guidance
Sampling guided away from self-collisions
q → surrogate → score → ∇ guidance
MLP surrogate · AUROC 0.968

Core Idea

Joint limits are enforced by construction, while self-collision is handled through learned diffusion guidance.

Text-Conditioned Humanoid Pose Generation with Physical Constraints

| Results & Current Extension

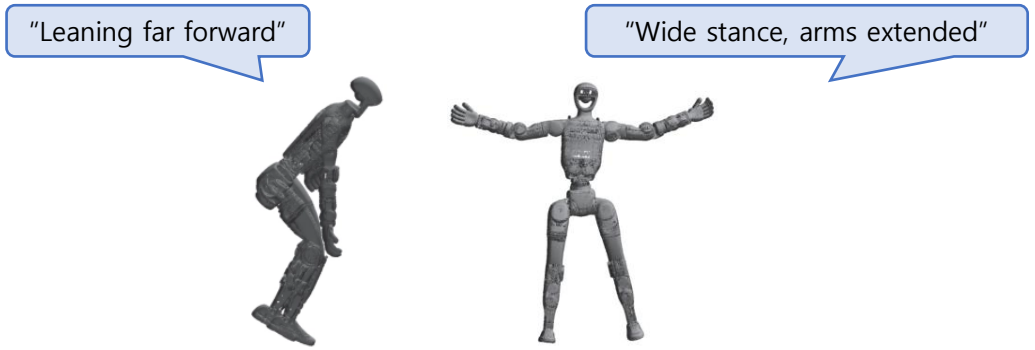
1. Physical Constraint Satisfaction

→ Joint-Limit Violations 0%
→ Self-Collision Rate 44.52% → 3.86%

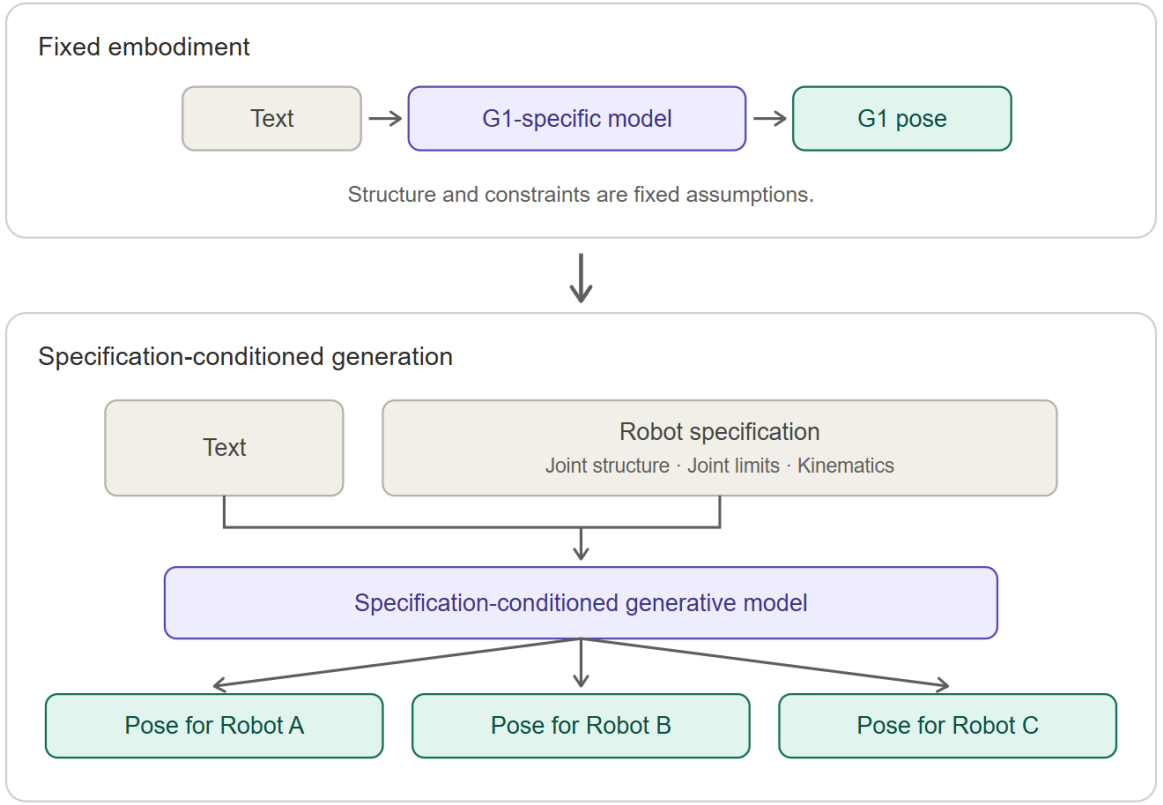
2. Does constraint Guidance Hurt Generation Quality?

Metric	Diffusion	Proposed
Laterality ↑	0.991	0.982
Direction ↑	0.762	0.762
Diversity ↑	10.93°	10.17°
Self-Collision ↓	44.52%	3.86%

→ Physical validity improves dramatically, while text control and diversity are largely preserved.



3. Current Research Extension



Embodiment structure and constraints become explicit model inputs

Text-Driven 2D Human Pose Generation via Diffusion in Directional Angle Space

Bomi Kwon, Ki Yong Lee

| Redesigning the generative output space to preserve skeletal structure by construction

Motivation

→ Coordinate-based generation predicts joints independently, making bone-length and geometric consistency difficult to preserve.

Research Question :

→ Can a structure-aware representation make diffusion-based pose generation more stable and controllable?

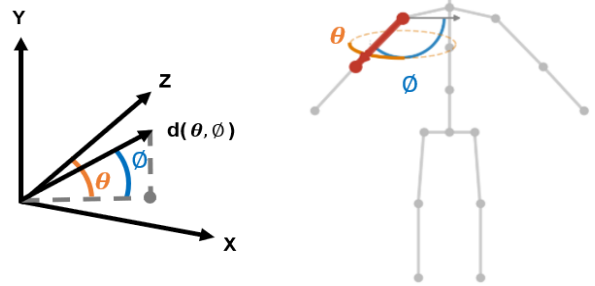
Key Idea

Directional Angle Representation

11 bones x 2 directional angles → 22D pose vector

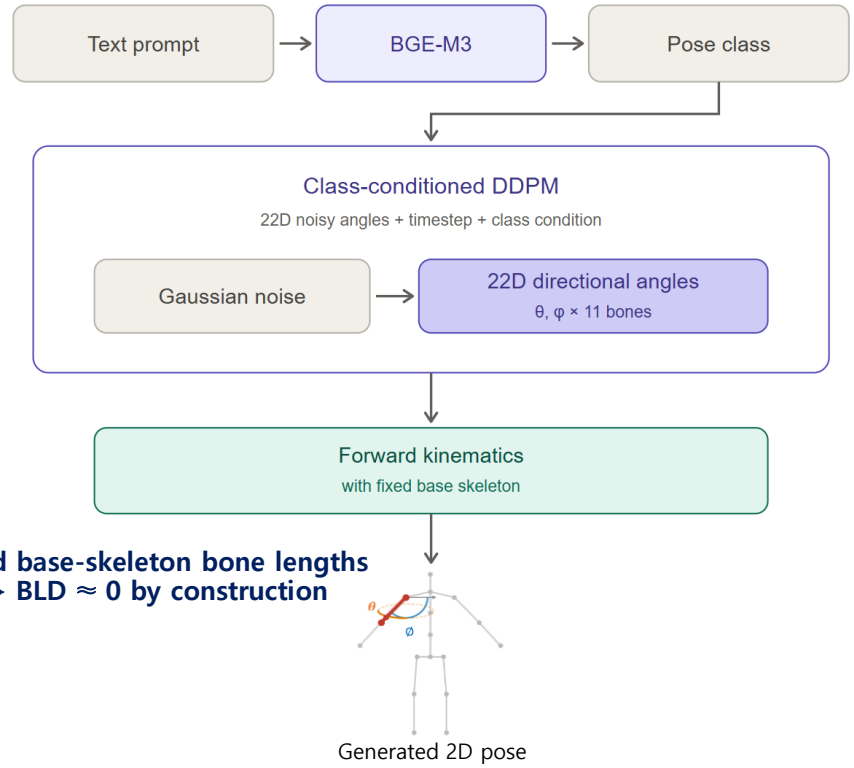
Each bone is represented by its direction, rather than by independently generated joint coordinates.

(A) Angle Definition **(B) Application to skeleton bones**



Metric	Coordinate	Ours
Semantic Consistency ↑	70.2%	**90.7%** + 20.5%p
FID ↓	8.69	**0.54**
Diversity	8.52	**9.01**
Bone-length deviation ↓	10.647	**0.000**

* FID represents distance to the real pose distribution



Contribution

Redesigned the diffusion output space with directional angles to improve semantic consistency while preserving skeletal structure by construction.

Selected Engineering & Research Projects

1 . Large-Scale Multimodal Product Matching & Retrieval

Industry-Academic Collaboration 2025 - present

- Visual product matching with Dreamsim / embedding models
- Semantic retrieval over 8M+ products
- Qdrant-based vector search

2. Agentic Retrieval System with LangGraph & MCP

Industry-Academic Collaboration 2025 - present

- Integrated product/image/semantic search into MCP tool interface
- Built LangGraph-based multi-step retrieval workflow
- Enabled LLM clients to call retrieval tools and perform follow-up searches

3. LLM-Based Graph Retrieval System 2025

- Natural language → Cypher query generation
- Neo4j-based large-scale graph retrieval
- Graph generation, metadata storage, and visualization

Projects focus : building scalable retrieval systems that connect generative AI, structured data, and tool-using agents.

Selected Publications & Recognition

Selected Publications
CIKM 2026 Accepted Short Paper
Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion
Bomi Kwon, Ki Yong Lee

The Transactions of the Korea Information Processing Society · 2026
Text-Driven 2D Human Pose Generation via Diffusion in Directional Angle Space
Bomi Kwon, Ki Yong Lee

Ongoing Research
Text-Conditioned Humanoid Pose Generation with Physical Constraints
Constraint-aware diffusion & embodiment-generalizable generation

---

Honors & Awards
WISET Engineering Research Team Program for Women Graduate Students
Research Principal Investigator · 2026
Selected as one of 50 research teams nationwide

Sookmyung Leadership Activity Scholarship · 2023

Selected Presentations
Diffusion-based Skeleton Pose Editing
Spring Academic Conference 2026 Oral Presentation

A Graph Generation Model for Text-based Graph Editing
KSC 2025 Oral Presentation

Development of a Large-Scale Graph Retrieval System based on LLM and Graph Database Providing a Natural Language Interface
KCC 2025 Presentation

Thank You

Bomi Kwon
M.S. Candidate in Computer Science
Sookmyung Women's University

https://kwonbomi.github.io/

bomi9970@sookmyung.ac.kr